

Binary Leibniz Algebras

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Abstract—An algebra is called a binary Leibniz algebra if each of its two-generated subalgebras is a Leibniz algebra. In the present paper, we give a description of binary Leibniz algebras in terms of identities. As a consequence, we show that the variety of binary Leibniz algebras is not Schreier and that the freedom theorem fails to hold for this variety.

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1. INTRODUCTION

For a class of algebras $\mathcal{A}$, denote by $\mathcal{A}_1$ the class of algebras in which every singly generated algebra belongs to the class $\mathcal{A}$. We similarly define $\mathcal{A}_2$ as the class of algebras in which every two-generated algebra belongs to the class $\mathcal{A}$.

An algebra A is called a *mono $\mathcal{A}$ -algebra* if $A \in \mathcal{A}_1$, and a *binary $\mathcal{A}$ -algebra* if $A \in \mathcal{A}_2$.

For example, let $\mathcal{A} = \mathcal{A}s$ be the class of associative algebras, i.e., the class of algebras defined by the associative identity

$$(a, b, c) = (ab)c - a(bc) = 0.$$

By Artin's theorem [1], the class of binary associative algebras coincides with the class of alternative algebras, i.e., $\mathcal{A}s_2$ is defined by the polynomial identities

$$(a, b, c) = -(b, a, c) = -(a, c, b).$$

It follows from Albert's theorem [2], [3] that, over a field of characteristic zero, the class of mono associative algebras coincides with the class of power-associative algebras, i.e., $\mathcal{A}s_1$ is defined by the following identities:

$$(a, a, a) = 0, \quad (aa, a, a) = 0.$$

Another example concerns the class of Lie algebras $\mathcal{A} = \mathcal{L}ie$, i.e., the class defined by the identities

$$ab + ba = 0, \quad \text{Jac}(a, b, c) = 0,$$

where $\text{Jac}(a, b, c) = (ab)c + (bc)a + (ca)b$ is the Jacobian of the elements a, b, c . In this case, the class of binary Lie algebras $\mathcal{L}ie_2$ over a field of characteristic distinct from two is determined by the following polynomial identities [3]:

$$ab + ba = 0, \quad \text{Jac}(a, b, ab) = 0.$$

Note that the class of mono Lie algebras $\mathcal{L}ie_1$ coincides with the class of anticommutative algebras.

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An algebra is said to be *right alternative* if the identity

$$(a, b, b) = 0$$

holds in the algebra. A right alternative algebra is said to be a $(-1, 1)$ -algebra if the following identity holds in the algebra:

$$S(a, b, c) = 0, \quad \text{where} \quad S(a, b, c) = (a, b, c) + (b, c, a) + (c, a, b).$$

In [4], the defining identities of the class of binary $(-1, 1)$ -algebras over a field of characteristic zero were described. An algebra is a binary $(-1, 1)$ -algebra if and only if it satisfies the following two identities:

$$(a, b, b) = 0, \quad S(ab, a, b) = 0.$$

Recall that an algebra is called a (left) *Leibniz algebra* if it satisfies the following identity :

$$(ab)c = a(bc) - b(ac).$$

The defining identities of the class of mono Leibniz algebras over an infinite field were found in [5]. An algebra is a mono Leibniz algebra if and only if the following identities hold in it:

$$(aa)a = 0, \tag{1.1}$$

$$(aa)(aa) = 0. \tag{1.2}$$

In the present paper, we describe the defining identities of the variety of the binary Leibniz algebras. Before formulating our result, we introduce some notation.

Let $\mathbf{K}$ be a field whose characteristic differs from 2, and let A be an algebra over $\mathbf{K}$. For every $a, b, c \in A$, write

$$\langle a, b, c \rangle = (ab)c - a(bc) + b(ac).$$

It is clear that an algebra with the identity $\langle a, b, c \rangle = 0$ is a Leibniz algebra. In the case of anticommutative algebras, recall that

$$\langle a, b, c \rangle = \text{Jac}(a, b, c).$$

Theorem 1. *An algebra A over $\mathbf{K}$ is a binary Leibniz algebra if and only if the following identities hold in A :*

$$\langle a, a, b \rangle = 0, \tag{1.3}$$

$$\langle a, b, a \rangle = 0, \tag{1.4}$$

$$\langle a, b, ab \rangle = 0. \tag{1.5}$$

As is well known, every Lie algebra is a Mal'tsev algebra, and every Mal'tsev algebra is a binary Lie algebra. However, there is a binary Lie algebra that is not a Mal'tsev algebra, and there is a Mal'tsev algebra that is not a Lie algebra; see, for example, [6]. Since the Leibniz algebras are noncommutative generalizations of Lie algebras, it follows that every binary Lie algebra is a binary Leibniz algebra. By Theorem 1, every Leibniz algebra is a binary Leibniz algebra and every binary Leibniz algebra is a mono Leibniz algebra. Moreover, by (1.1) and (1.2), every mono Lie algebra is a mono Leibniz algebra. The inclusion diagram looks as follows:

$$\begin{array}{ccccccc} \mathcal{L}ie & \subset & \mathcal{M}alc & \subset & \mathcal{L}ie_2 & \subset & \mathcal{L}ie_1 \\ \cap & & & & \cap & & \cap \\ \mathcal{L}eib & \subset & & \subset & \mathcal{L}eib_2 & \subset & \mathcal{L}eib_1 \end{array}$$

Note that all inclusions indicated above are strict. Below we provide examples to confirm these facts.

Example 1 (non-Leibniz but a binary Leibniz algebra). Let $A = \{e_1, e_2, e_3, e_4\}$ be a four-dimensional anticommutative algebra with the multiplication table

$$e_1e_2 = e_3, \quad e_1e_4 = e_1, \quad e_2e_4 = e_2, \quad e_3e_4 = -e_3$$

where the zero products are omitted. In this case, A is a non-Lie Mal'tsev algebra [6]; hence A is a binary Leibniz algebra. However, A is not a Leibniz algebra, because $\langle e_1, e_2, e_4 \rangle = -3e_3 \neq 0$.

Example 2 (not a binary Lie algebra, but a binary Leibniz algebra). Let B be a free singly generated nilpotent algebra of index 3. Since B is not anticommutative, it is not a binary Lie algebra. On the other hand, B is a Leibniz algebra; hence, it is a binary Leibniz algebra.

Example 3 (not mono Lie, non-Leibniz, but a binary Leibniz algebra). The direct sum $A \oplus B$ of the two algebras defined above gives an example of a binary Leibniz algebra which is neither a Leibniz algebra nor a binary Lie algebra.

Example 4 (not mono Lie but a mono Leibniz algebra). Let $C = \{e_1, e_2, e_3, e_4\}$ be a four-dimensional not anticommutative algebra with the multiplication table

$$e_1e_1 = e_4e_4 = e_2, \quad e_1e_2 = e_3,$$

where the zero products are omitted. Then C is a mono Leibniz algebra but is not a binary Leibniz algebra, because $\langle e_1 + e_4, e_4, e_1 + e_4 \rangle = -e_3 \neq 0$.

Example 5 (not a binary Lie algebra that is mono Lie). Let $D = \{e_1, e_2, e_3\}$ be a three-dimensional anticommutative algebra with the multiplication table

$$e_1e_2 = e_3, \quad e_1e_3 = e_1, \quad e_2e_3 = e_2,$$

where the zero products are omitted. Then D is not a binary Lie algebra, because

$$\text{Jac}(e_1, e_2, e_1e_2) = -2e_3 \neq 0.$$

In 1930, Magnus [7] proved the famous freedom theorem (Freiheitssatz) for free groups, which is an important theorem of combinatorial group theory. An analog of this theorem was also studied for varieties of algebras. We say that a *freedom theorem* is valid for the algebras of a variety $\mathcal{V}$ if the following condition holds.

Freedom Theorem. Let $X = \{x_1, \dots, x_n\}$, and let $F(X)$ be a free algebra on the set X of a variety $\mathcal{V}$. If $f \in F(X)$ and $f \notin F(\{x_1, \dots, x_{n-1}\})$, then

$$\text{Id}(f) \cap F(\{x_1, \dots, x_{n-1}\}) = 0,$$

where $\text{Id}(f)$ is the ideal of $F(X)$ generated by the element f .

For example, freedom theorems are proved for Lie algebras [8], (anti)commutative algebras [9], and associative algebras [10]. In [11, Corollary 5], examples of varieties of algebras are given for which the freedom theorem fails to hold. For example, freedom theorems fail to hold for the algebras of the varieties $\mathcal{Malc}$, $\mathcal{Lie}_2$, and $\mathcal{Leib}$.

Corollary 1. The freedom theorem fails to hold for binary Leibniz algebras.

Proof. Let $F(\{x_1, x_2\})$ be the free binary Leibniz algebra in the generators x_1 and x_2 . Consider the polynomial $f = x_2^2 + x_1 \in F(\{x_1, x_2\})$. By identity (1.3), we obtain $fx_1 = x_1^2$ and $f \in F(\{x_1\})$. This implies that $\text{Id}(f) \cap F(\{x_1\}) \neq 0$. $\square$

A variety $\mathcal{V}$ is said to be *Schreier* if every subalgebra of a free algebra of the variety $\mathcal{V}$ is free. For example, $\mathcal{Lie}$ and $\mathcal{Lie}_1$ are Schreier varieties [12], [13], and the varieties $\mathcal{Leib}$ and $\mathcal{Leib}_1$ are not [5]. By this result of the paper [5] and by Theorem 1, we obtain the following corollary.

Corollary 2. $\mathcal{Leib}_2$ is not a Schreier variety.

In the proof of Theorem 1, we use the method of linearizing the identities. Let $a, b, f \in A$ and let $\Delta_a^1(b)$ be a mapping of partial linearization of A (see [1, Chap. 1]). Write $\Delta(f)$ instead of $(f)\Delta_a^1(b)$ and call this operation the *differential substitution* $a \rightarrow b$ into f . The precise meaning of $\Delta(f)$ will be clear from the context. Denote by $\deg(f)$ the degree of the element f .

In the proof of Theorem 1, we need a basis of the free Leibniz algebra of a given set. As was proved in [14], the set of right normalized bracket elements of every set forms a base of the free Leibniz algebra (from the left) of the given set.

2. PROOF OF THEOREM 1

Let A be an algebra with the identities (1.3), (1.4), and (1.5). The full linearization of these identities gives, respectively, the equivalent identities

$$\langle a, b, c \rangle + \langle b, a, c \rangle = 0, \quad (2.1)$$

$$\langle a, b, c \rangle + \langle c, b, a \rangle = 0, \quad (2.2)$$

$$\langle a, b, cd \rangle + \langle a, d, cb \rangle + \langle c, b, ad \rangle + \langle c, d, ab \rangle = 0. \quad (2.3)$$

It can readily be seen that, in view of identities (2.1) and (2.2), $\langle a, b, c \rangle$ is skew-symmetric with respect to a, b, c . It follows from identity (1.3) that $(aa)b = 0$. Hence

$$\langle a^2, b, c \rangle = 0. \quad (2.4)$$

Proof of Theorem 1. Since (1.3), (1.4), and (1.5) depend on two variables only, it follows that, over a field of arbitrary characteristic, every binary Leibniz algebra satisfies these identities.

Let $x, y \in A$, and let $\text{alg}\langle x, y \rangle$ be a two-generated subalgebra of A in x and y . For every $u, v, w \in \text{alg}\langle x, y \rangle$, we will prove that

$$\langle u, v, w \rangle = 0$$

using induction on the degree $\deg(\langle u, v, w \rangle)$. If $\deg(\langle u, v, w \rangle)$ is equal to 3 or 4, then our assertion is obvious by identities (1.3), (1.4), and (1.5).

Suppose that $\langle u, v, w \rangle = 0$ if $\deg(\langle u, v, w \rangle)$ does not exceed n . Let $\deg(\langle u, v, w \rangle) = n + 1$. If $\deg(w) \geq 2$, then, by the differential substitution $x \rightarrow w$ into $\langle u, v, x \rangle = 0$, we obtain

$$\Delta(\langle u, v, x \rangle) = \langle \Delta(u), v, x \rangle + \langle u, \Delta(v), x \rangle + \langle u, v, w \rangle = 0.$$

Thus, if we prove that all triples of the form $\langle u, v, x \rangle$ of degree $n + 1$ vanish in A , then this will prove that $\langle u, v, w \rangle = 0$.

We claim now that, in fact, $\langle u, v, x \rangle$ of degree $n + 1$ is a linear combination of triples of the form $\langle u, x, y \rangle$. Suppose that $\deg(v) \geq 2$ and consider the differential substitution $y \rightarrow v$ into $\langle u, y, x \rangle = 0$. Then we have

$$\Delta(\langle u, y, x \rangle) = \langle \Delta(u), y, x \rangle + \langle u, v, x \rangle + \langle u, y, \Delta(x) \rangle = 0.$$

This gives the following relation:

$$\langle u, v, x \rangle = -\langle \Delta(u), y, x \rangle.$$

Hence we are to prove that all triples of the form $\langle u, x, y \rangle$ vanish in A .

Suppose that $\deg(\langle u, x, y \rangle) = n$. The differential substitution $x \rightarrow xy$ into $\langle u, x, y \rangle$ implies the following relation:

$$\Delta(\langle u, x, y \rangle) = \langle \Delta(u), x, y \rangle + \langle u, \Delta(x), y \rangle + \langle u, x, \Delta(y) \rangle = 0.$$

This implies that

$$\langle \Delta(u), x, y \rangle + \langle u, xy, y \rangle = 0. \quad (2.5)$$

By the differential substitution $x \rightarrow u$ into $\langle xy, x, y \rangle$ we have

$$\Delta(\langle xy, x, y \rangle) = \langle uy, x, y \rangle + \langle xy, \Delta(x), y \rangle + \langle xy, x, \Delta(y) \rangle = 0.$$

Hence

$$\langle uy, x, y \rangle + \langle xy, u, y \rangle = 0. \quad (2.6)$$

Let us sum (2.5) and (2.6). Since $\langle \cdot, \cdot, \cdot \rangle$ is skew-symmetric, we have

$$\langle \Delta(u) + uy, x, y \rangle = 0. \quad (2.7)$$

We claim now that, for every $v \in A$, in the case of a differential substitution $x \rightarrow xy$ into u of degree $n - 2$, the following relation holds:

$$(\Delta(u) + uy)v = 2(uy)v. \quad (2.8)$$

We prove this by induction on the degree of u . If $u = xy$, then (2.8) holds. The validity of relation (2.8) in the case $u = yx$ can readily be proved using identity (2.1).

Suppose that our assertion holds for all u whose degree is less than $n - 2$. Let $\deg(u) = n - 2$ and $u = u_1u_2$. Since $\deg((u_1u_2)y) = n - 1$, it follows that, by the induction assumption of the first induction, we can write

$$(u_1u_2)y = u_1(u_2y) - u_2(u_1y).$$

Then, by (2.1) and by the induction assumption, we have

$$\begin{aligned} (\Delta(u) + uy)v &= (\Delta(u_1u_2) + (u_1u_2)y)v = (\Delta(u_1)u_2 + u_1\Delta(u_2) + (u_1u_2)y)v \\ &= (\Delta(u_1)u_2 + u_1\Delta(u_2) + u_1(u_2y) - u_2(u_1y))v \\ &= (\Delta(u_1)u_2 - \Delta(u_2)u_1 - (u_2y)u_1 + (u_1y)u_2)v \\ &= ((\Delta(u_1) + u_1y)u_2 - (\Delta(u_2) + u_2y)u_1)v = (2(u_1y)u_2 - 2(u_2y)u_1)v \\ &= 2((u_1y)u_2 - (u_2y)u_1)v = 2(u_1(u_2y) - u_2(u_1y))v \\ &= 2((u_1u_2)y)v = 2(uy)v. \end{aligned}$$

Thus, by (2.7), (2.8), and the full linearization of identity (2.4), we can obtain

$$0 = \langle \Delta(u) + uy, x, y \rangle = \langle 2uy, x, y \rangle = 2\langle uy, x, y \rangle = -2\langle yu, x, y \rangle.$$

Since $\mathbf{K}$ is a field whose characteristic differs from two, we have

$$\langle yu, x, y \rangle = 0. \quad (2.9)$$

Further, we can similarly prove that

$$\langle xv, x, y \rangle = 0 \quad (2.10)$$

for all v of degree $n - 2$. Using the basis of the free Leibniz algebra, we see that every element $w \in A$ of degree $n - 1$ can be expressed as a linear combination of elements of the form yu and xv . Thus, by (2.9) and (2.10), for all $w \in A$ we have

$$\langle w, x, y \rangle = 0.$$

It remains to note that all identities (1.3), (1.4), and (1.5) are independent. The independence of identities (1.3) and (1.4) is obvious.

Consider the three-dimensional algebra D from Example 5. Then identities (1.3) and (1.4) hold in the algebra D . However, this algebra does not satisfy identity (1.5), because $\langle e_1, e_2, e_1e_2 \rangle = -2e_3 \neq 0$. This implies that identity (1.5) is not a consequence of identities (1.3) and (1.4), and hence the identities (1.3), (1.4), and (1.5) are independent. This completes the proof of the theorem. $\square$

3. REMARKS AND OPEN QUESTIONS

3.1. Note that there is a gap between $\mathcal{L}ie$ and $\mathcal{L}ie_2$:

$$\begin{array}{ccccccc} \mathcal{L}ie & \subset & \mathcal{M}al & \subset & \mathcal{L}ie_2 & \subset & \mathcal{L}ie_1 \\ \cap & & \cap & & \cap & & \cap \\ \mathcal{L}ie & \subset & ? & \subset & \mathcal{L}ie_2 & \subset & \mathcal{L}ie_1 \end{array}$$

It would be of interest to find a Leibniz analog of Mal'tsev algebras (if it exists) that would make our diagram full.

3.2. In [15], Bremner, Peresi, and Sánchez-Ortega defined Mal'tsev dialgebras over a field whose characteristic differs from two. A dialgebra is said to be (left) *Mal'tsev dialgebra* if it satisfies the following identities:

$$\begin{aligned} (ab)c + (ba)c &= 0, \\ a(b(cd)) - b(c(ad)) - c((ab)d) - (ac)(bd) - (a(bc))d &= 0. \end{aligned}$$

The first identity is called the *left anticommutativity*, and the second the (left) *di-Mal'tsev* identity. Let di-Malc be the class of Mal'tsev dialgebras. Since a Mal'tsev dialgebra is a “noncommutative” analog of a Mal'tsev algebra, we have

$$\text{Malc} \subset \text{di-Malc}.$$

It can readily be proved that

$$\mathcal{L}\text{eib} \subset \text{di-Malc} \quad \text{and} \quad \text{di-Malc} \subset \text{di-L}\mathcal{e}\text{ib}_1.$$

However,

$$\text{di-Malc} \not\subset \mathcal{L}\text{eib}_2.$$

To see this, consider the algebra C from Example 4, i.e., the algebra with the multiplication table

$$e_1e_1 = e_4e_4 = e_2, \quad e_1e_2 = e_3,$$

where the zero products are omitted. Note that e_2 and e_3 are left central elements of C , i.e.,

$$e_2a = 0 \quad \text{and} \quad e_3a = 0$$

for every $a \in C$, and C^2 is spanned by e_2 and e_3 . It can be proved that C is a nilpotent algebra with nilpotency index 4. Then C satisfies the left anticommutativity and the di-Mal'tsev identity, and hence C is a Mal'tsev dialgebra. By Example 4, C is not a binary Leibniz algebra. Thus, a Mal'tsev dialgebra need not be a binary Leibniz algebra. Hence the class of Mal'tsev dialgebras cannot fill the gap in our diagram.

3.3. Another interesting class of algebras containing all Leibniz and Mal'tsev algebras is the class of binary Lie dialgebras. Let di-Lie_2 be the class of binary Lie dialgebras. To derive the defining identities of di-Lie_2 , one can use the general algorithm described in [16]. Then a dialgebra is called a *binary Lie dialgebra* if it satisfies the following identity:

$$a(b(cd)) + a(c(bd)) - ((ac)b)d - ((ab)c)d - b(a(cd)) - c(a(bd)) + b((ca)d) + c((ba)d) = 0.$$

Let us rewrite the di-Mal'tsev identity as follows:

$$\begin{aligned} c(a(bd)) - a(b(cd)) - b((ca)d) - (cb)(ad) - (c(ab))d &= 0, \\ b(a(cd)) - a(c(bd)) - c((ba)d) - (bc)(ad) - (b(ac))d &= 0. \end{aligned}$$

To derive the identity of a binary Lie dialgebra, we apply the identity of left anticommutativity to the sum of the two last Mal'tsev identities. Then we have

$$\text{di-Malc} \subset \text{di-Lie}_2$$

and, therefore,

$$\mathcal{L}\text{eib} \subset \text{di-Lie}_2 \quad \text{and} \quad \text{Malc} \subset \text{di-Lie}_2.$$

Since $\text{di-Malc} \not\subset \mathcal{L}\text{eib}_2$, we have

$$\text{di-Lie}_2 \not\subset \mathcal{L}\text{eib}_2.$$

Hence the class of binary Lie dialgebras cannot fill the gap in our diagram.

3.4. There are no simple non-Lie algebras in the class of Leibniz algebras; however, the class of binary Lie algebras contains a simple non-Lie algebra, which is in fact a Mal'tsev algebra [17]. Hence the class of binary Leibniz algebras contains simple non-Lie algebras. It would be also of interest to find a non-binary Lie algebra which is a binary Leibniz algebra (if it exists).

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